

CSE 2216 - Application Development Lab

Project Report



Project Name

Learn Sphere

Submitted By

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28th Batch

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1. Introduction

1.1 About the App

LearnSphere is an interactive educational platform that enables students to enroll in courses, attend live classes, participate in quizzes, and engage with instructors and peers through chat.

1.2 Motivation

In today's fast-paced world, access to quality education should not be limited by location, time, or financial constraints. Traditional learning methods often lack flexibility, interactivity, and real-time engagement, making it difficult for students to grasp concepts effectively.

The **LearnSphere** app was developed to address these challenges by providing a dynamic and interactive learning environment. The motivation behind building this platform includes:

- **Enhancing Online Education:** Many existing platforms are either too complex or lack essential features like real-time interaction, making learning less engaging. Our app simplifies the process while ensuring a seamless experience for both students and instructors.
- **Bridging the Teacher-Student Gap:** Traditional online courses often feel one-sided, where students passively consume content. This app encourages active participation through live classes, quizzes, and interactive chat features.
- **Encouraging Gamification & Engagement:** Learning is more effective when students stay motivated. Features like leaderboards and quizzes help foster a competitive yet fun learning experience.
- **Facilitating Secure and Easy Transactions:** Many e-learning platforms require complicated payment processes. Our integrated local payment gateway ensures hassle-free course enrollment.
- **Providing a One-Stop Learning Solution:** From live classes and quizzes to file sharing and chat, our app offers a comprehensive learning ecosystem where everything is accessible in one place.

1.3 Features

- **User Authentication:** Secure login and role-based access control (students & instructors).
- **Live Classes:** Real-time virtual classroom experience.
- **Interactive Quizzes:** Engage students with quizzes launched by instructors.
- **Leaderboard:** Track student performance and display rankings.
- **Payment Gateway:** Secure transactions for course enrollment.
- **File Uploads:** Support for various file types, including PDFs, videos, and documents.

- **Chat System:** Real-time messaging between students and instructors.

1.4 Tools, Technologies and Frameworks Used

1. **Mobile Development:** Flutter (Dart) for Android.
2. **Backend & Database:** Firebase, Firestore, FirebaseAuth for authentication and data storage
3. **Live Class:** Zigo Cloud

1.5 Individual Responsibilities

1.4.1 Abu Horaira

1. Student and Instructor Pages designing
2. Database Integration and handling
3. Functionality between Screens and Updates
4. Application Structure and File management

1.4.2 Shraban Karmoker Avi

1. Live Class implementation and updating features
2. Payment Gateway implementation
3. Database Connectivity
4. Documentations

1.4.3 Showkoth Osman Shova

1. Chat Application
2. UI upgrading
3. LeaderBoard

1.4.4 Adib Ahsan

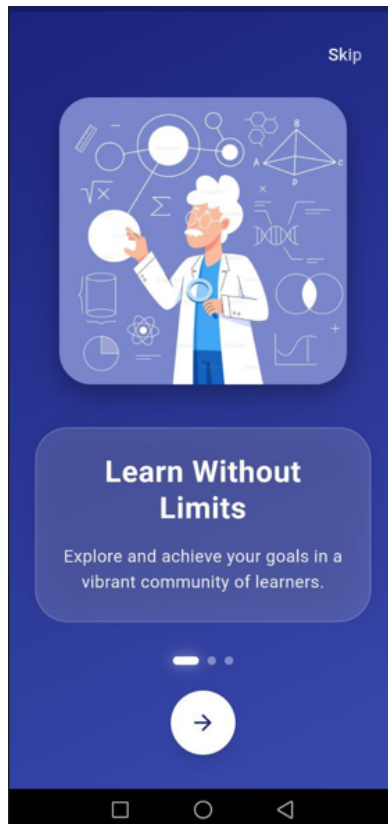
1. Database Design
2. Authorization
3. Interactive Quiz Section
4. App Testing

2. Design and Implementation

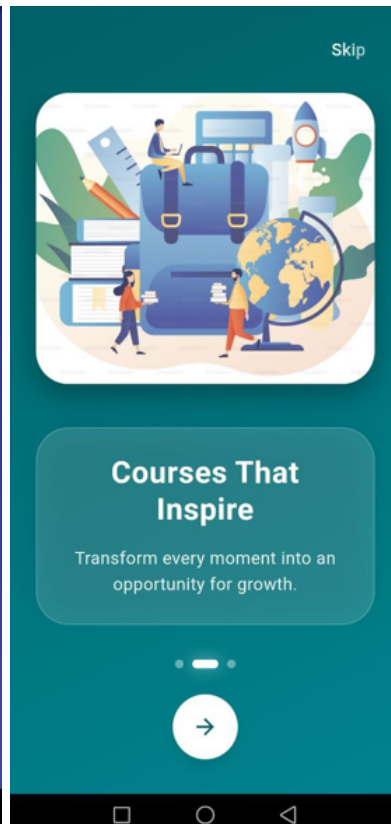
2.1 UI Developed (Screenshots)

- Onboarding pages

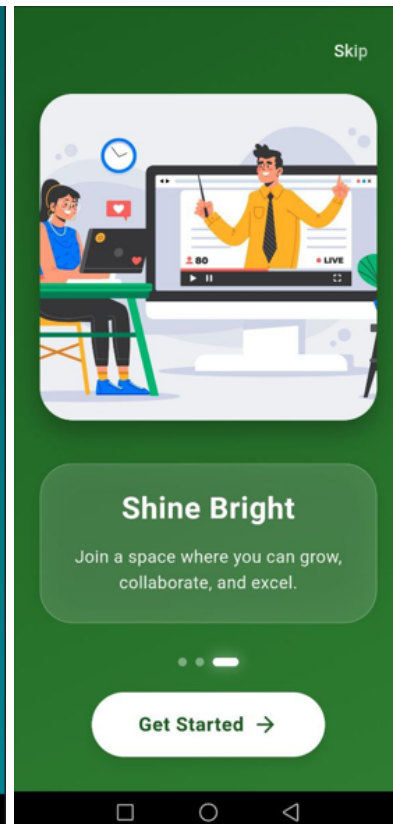
Page1



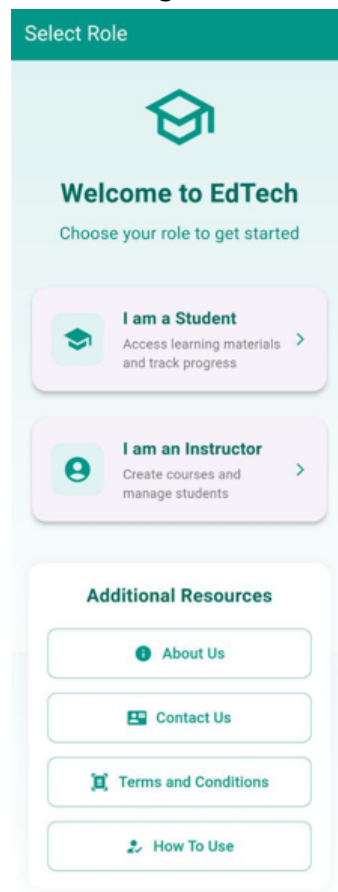
Page2



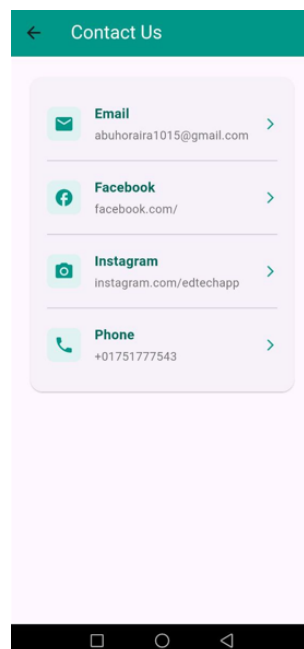
Page3



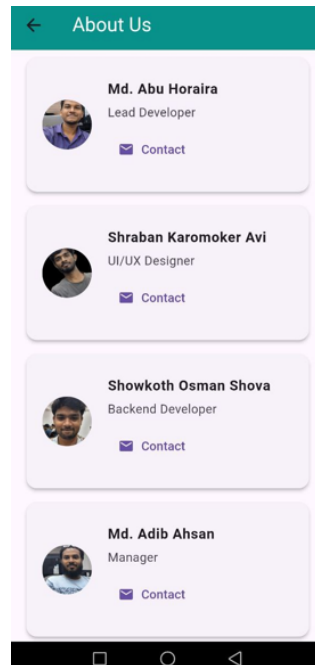
- **Role Selection Page**



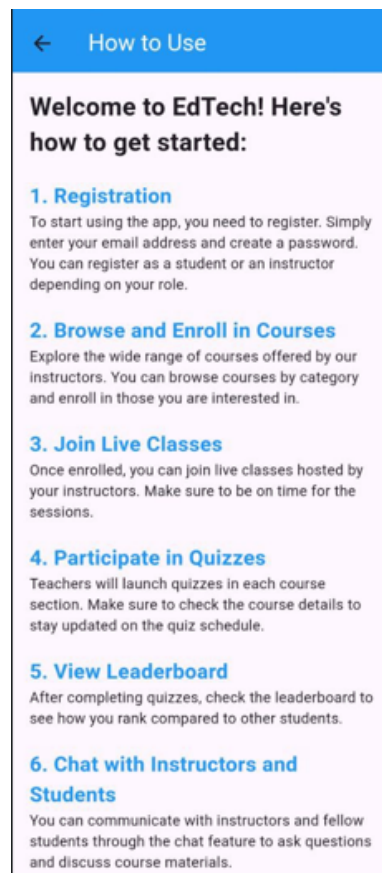
- **Contact Us**



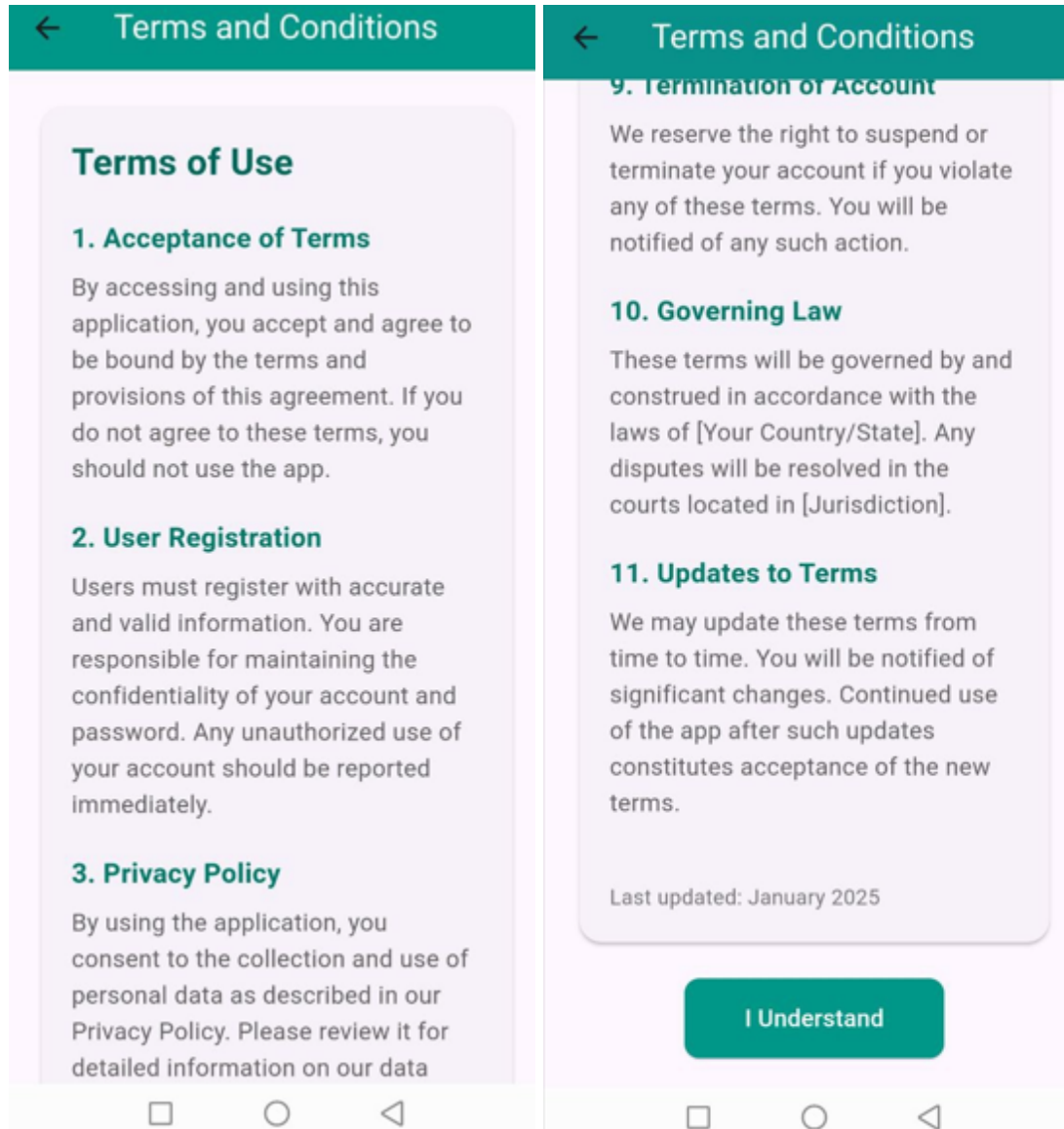
- **About Us**



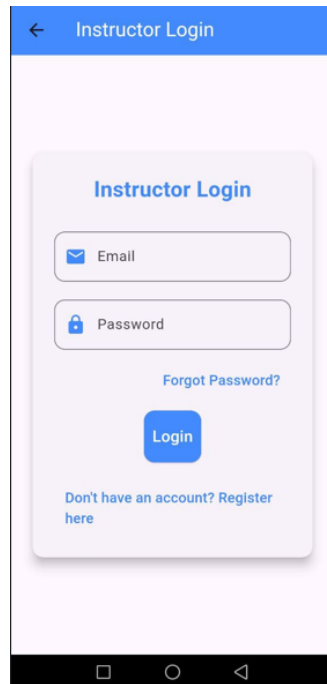
- **How To Use**



- **Terms And Conditions**

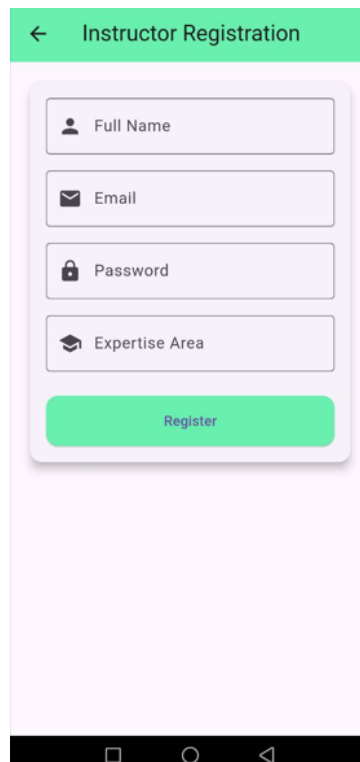


- **Instructor Pages**
 - **Login**



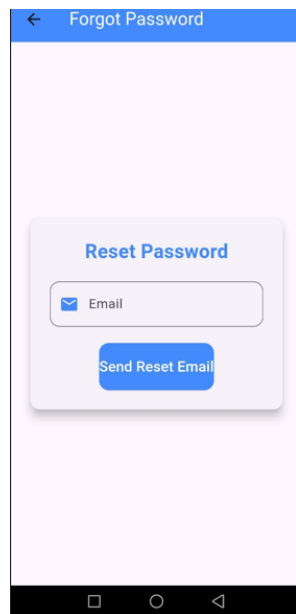
The image shows a mobile app screen for "Instructor Login". At the top is a blue header bar with a back arrow and the text "Instructor Login". Below this is a light purple rounded rectangle containing the login form. The form has two input fields: "Email" with an envelope icon and "Password" with a lock icon. Below the password field is a link "Forgot Password?". At the bottom of the form is a blue "Login" button. Below the button is a link "Don't have an account? Register here". The screen has a black Android navigation bar at the bottom.

- **Registration**

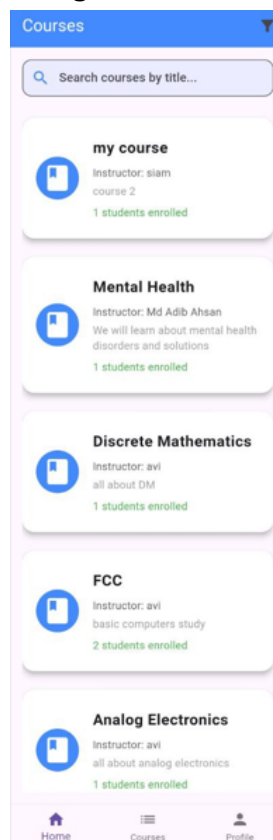


The image shows a mobile app screen for "Instructor Registration". At the top is a green header bar with a back arrow and the text "Instructor Registration". Below this is a light purple rounded rectangle containing the registration form. The form has four input fields: "Full Name" with a person icon, "Email" with an envelope icon, "Password" with a lock icon, and "Expertise Area" with a graduation cap icon. Below the form is a green "Register" button. The screen has a black Android navigation bar at the bottom.

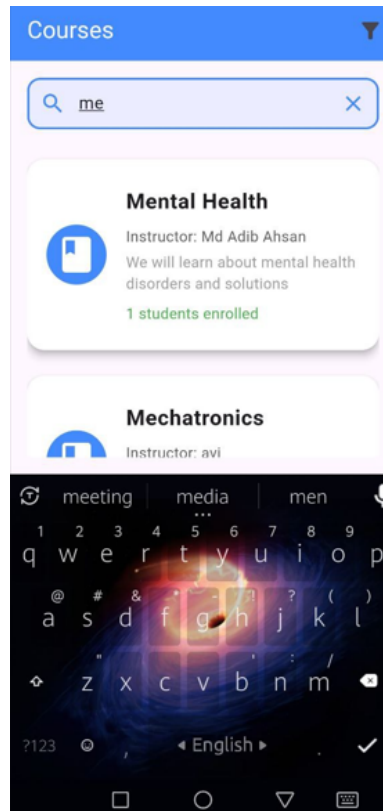
- **Forgot Password**



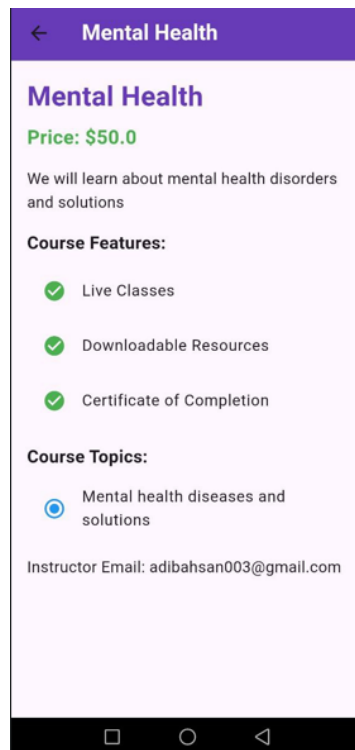
- **Courses Page**



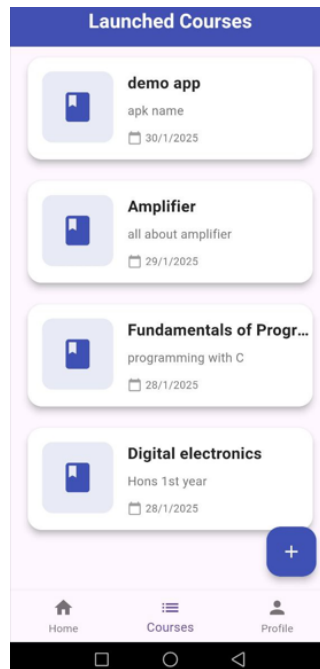
○ **Search Course**



○ **View Other instructor course**

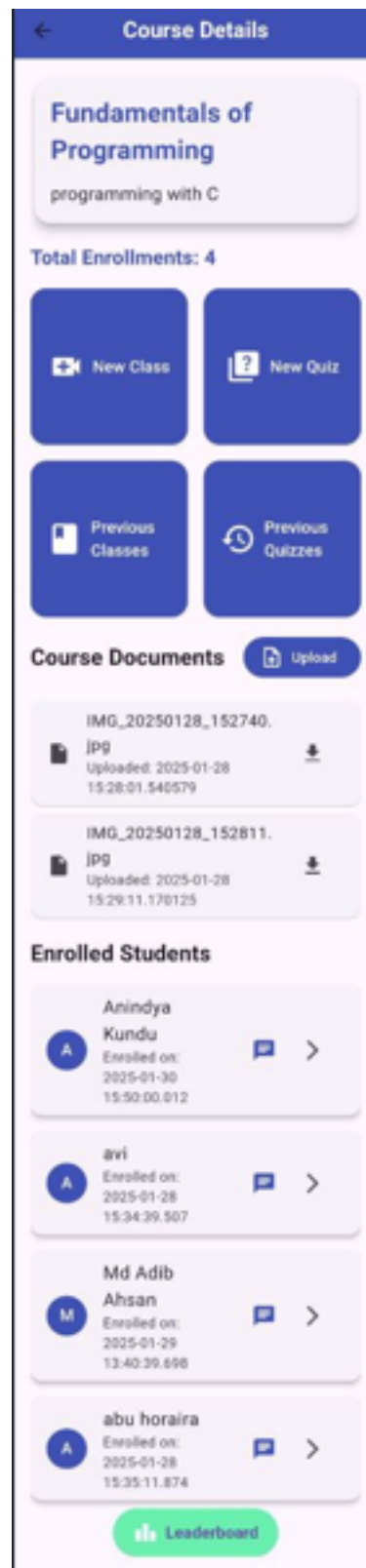


- **Launched Courses**



- **Launch New Course**

- Course Details



- **Launch Class**

A screenshot of a mobile application interface for launching a new class. The screen has a blue header bar with a back arrow and the text "Launch New Class". Below the header, there are four text input fields stacked vertically: "Class Title", "Class Description", "Class Time (e.g., 10:00 AM)", and "Maximum Students". At the bottom of the form is a blue button with the text "Create Class". The bottom of the screen shows a black navigation bar with three white icons: a square, a circle, and a triangle.

- **View Previous Class**

A screenshot of a mobile application interface for viewing previous classes. The screen has a blue header bar with a back arrow and the text "Previous Classes". Below the header, there are two class cards stacked vertically. Each card has a title, a description, a time, a maximum number of students, a status icon, enrollment information, a date, and a "Start Meeting" button. The first card is titled "Variable" and the second is titled "Data type".

Class Title	Description	Time	Max Students	Enrolled	Date	Action
Variable	Description: all about variables	Time: 9	Max Students: 14	Enrolled: 0 students	Date: 28/1/2025 at 15:25	Start Meeting
Data type	Description: all about data thpes	Time: 11	Max Students: 34	Enrolled: 0 students	Date: 28/1/2025 at 15:25	Start Meeting

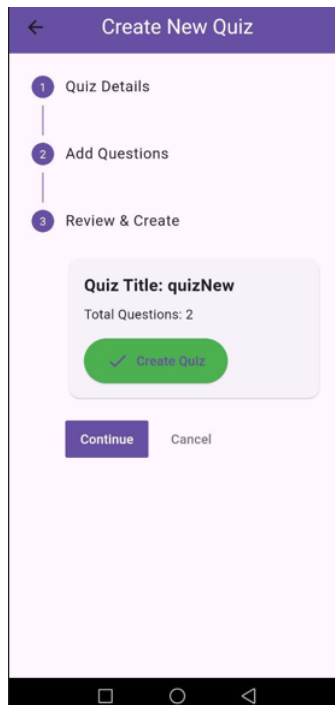
- **Launch Quiz**
 - **Quiz title**

The screenshot shows the 'Create New Quiz' app interface. At the top is a purple header with a back arrow and the text 'Create New Quiz'. Below the header is a vertical progress indicator with three steps: 1. Quiz Details (active), 2. Add Questions, and 3. Review & Create. The 'Quiz Details' step contains a text input field labeled 'Quiz Title' with a 'T' icon. Below this is a 'Quiz Information' box showing 'Course ID: course2'. At the bottom of the step are two buttons: 'Continue' (purple) and 'Cancel' (grey). The app is running on an Android device, as indicated by the navigation bar at the bottom.

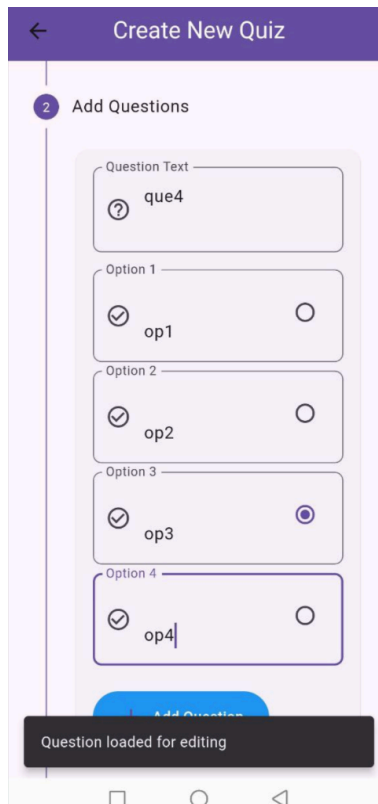
- **Questions and Right answer selection**

The screenshot shows the 'Create New Quiz' app interface at the 'Add Questions' step. The purple header remains at the top. The progress indicator shows step 2, 'Add Questions', as the active step. The main area contains a 'Question Text' input field with a question mark icon and the text 'que2'. Below this are four 'Option' input fields, each with a checkmark icon, the option text, and a radio button. The options are: 'op1', 'op2', 'op3', and 'op4'. The 'op4' radio button is selected. Below the options is a blue button with a plus icon and the text 'Add Question'. At the bottom, there is a section titled 'Added Questions:' which shows a list of added questions, currently containing 'Q1: que1'. The app is running on an Android device, as indicated by the navigation bar at the bottom.

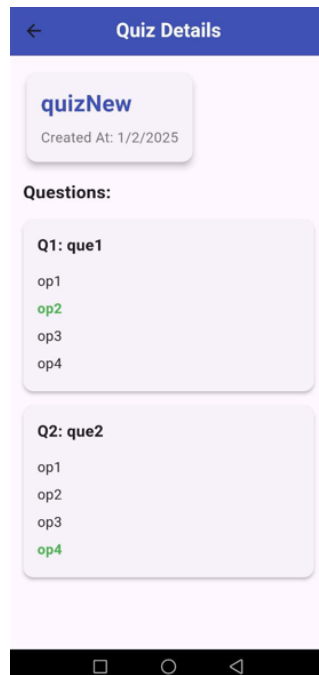
■ Create Quiz



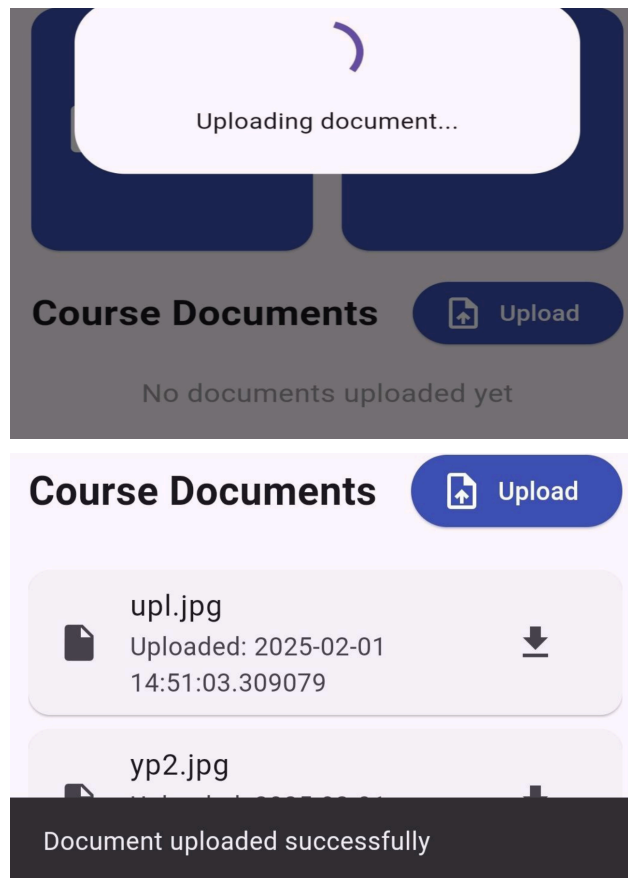
■ Edit Quiz



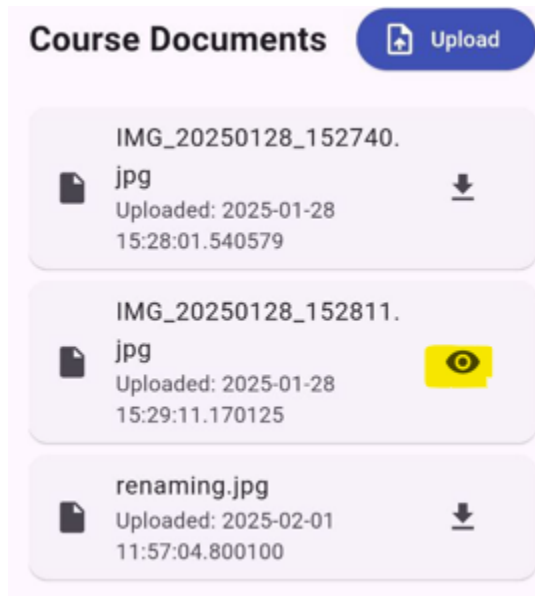
- Previous Quiz



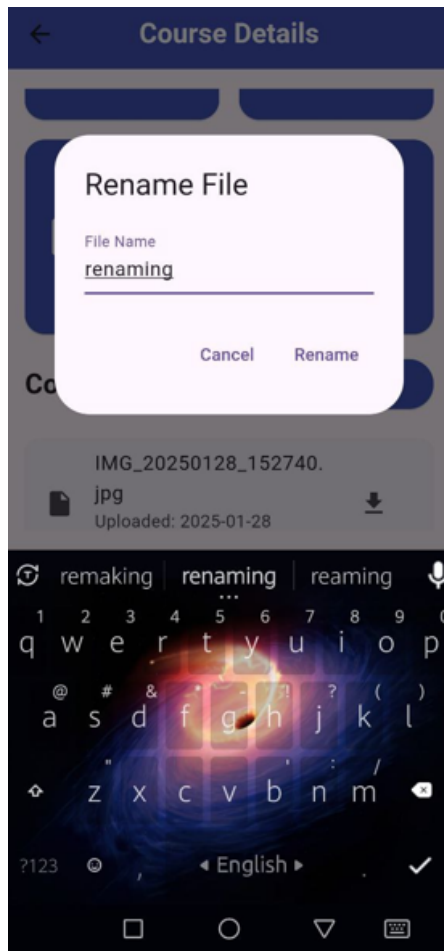
- Upload Documents



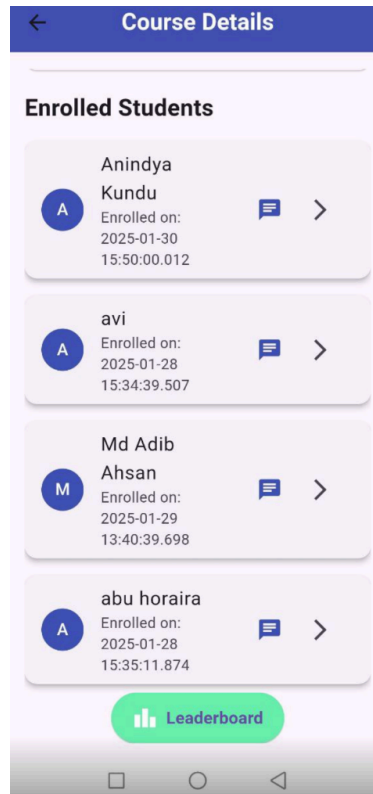
- **View Documents**



- **Rename Document**



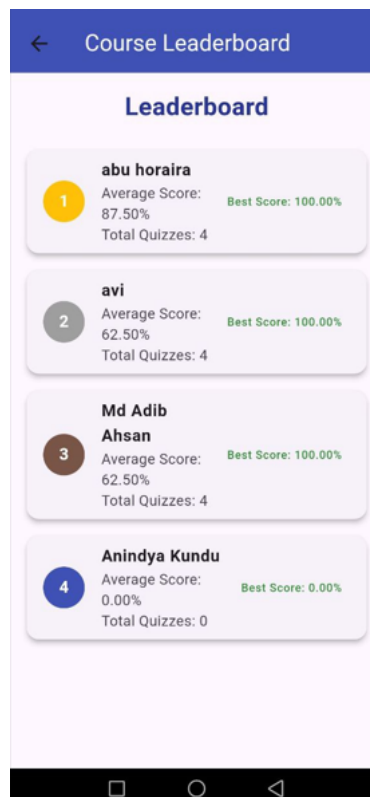
- **Enrolled Students**



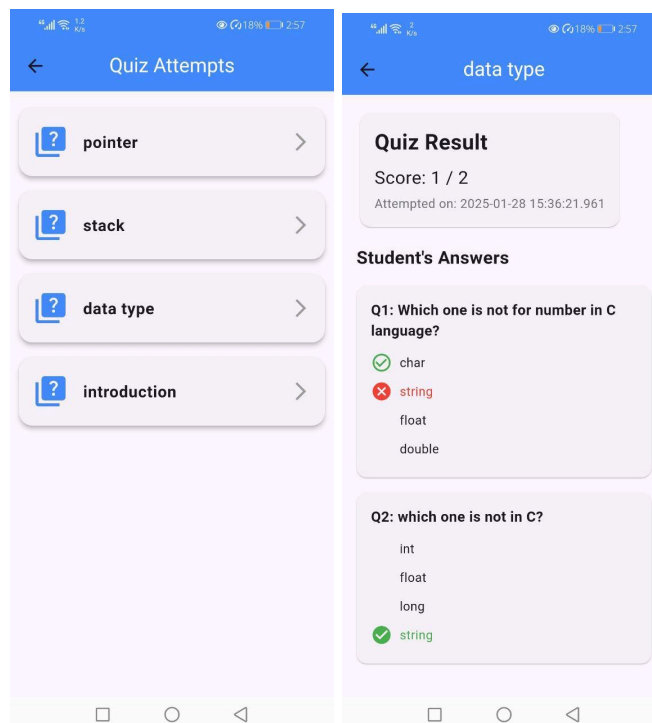
- **Chat with students**



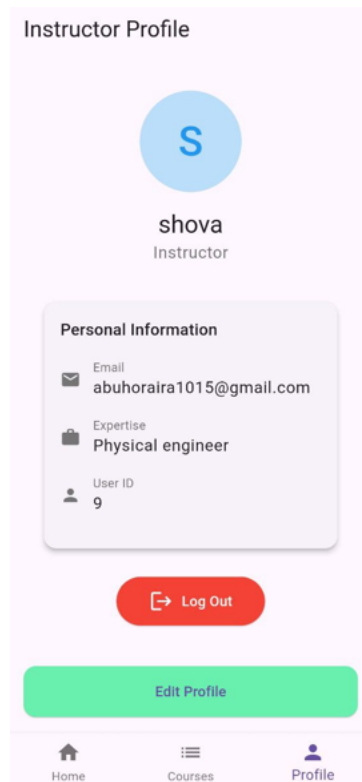
○ **Leaderboard**



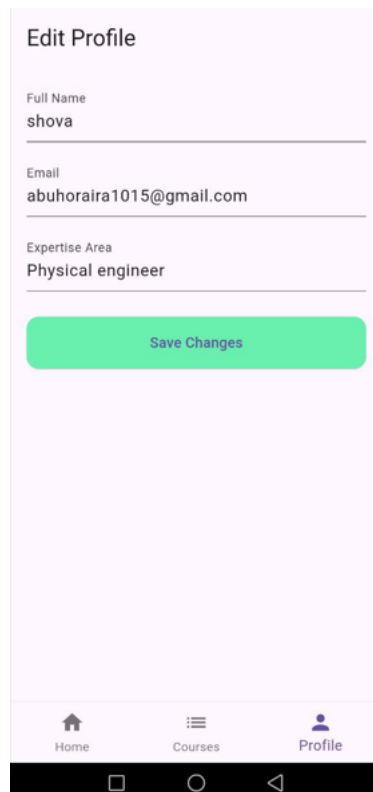
○ **Student Quiz attempts**



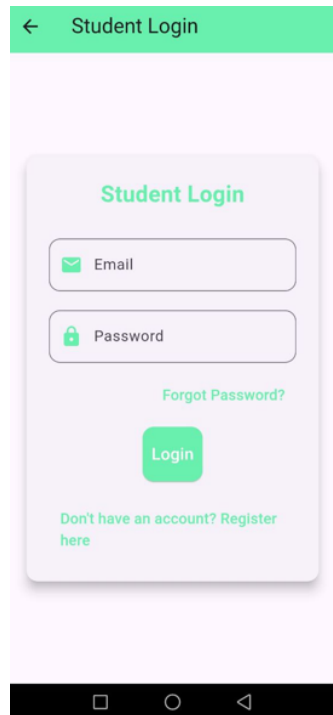
- **Profile**



- **Edit Profile**

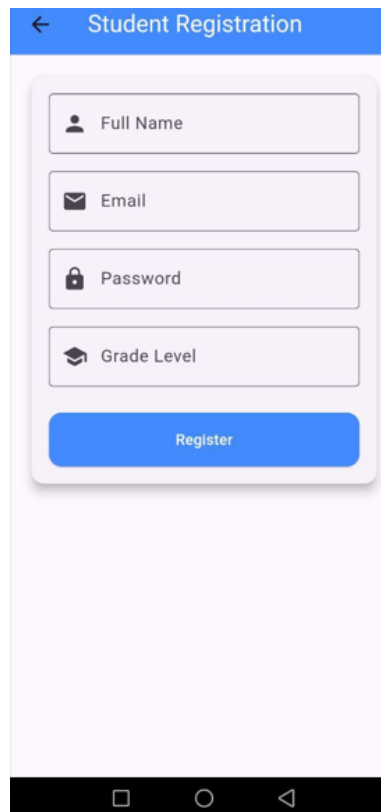


- **Student Pages**
 - **Login**



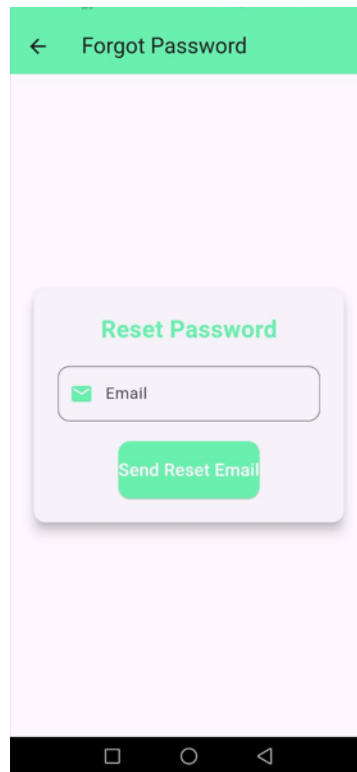
A mobile app screen for student login. At the top is a green header bar with a white back arrow and the text "Student Login". The main content is a light purple rounded rectangle with a shadow. Inside, the title "Student Login" is in green. Below it are two input fields: "Email" with a green checkmark icon and "Password" with a green lock icon. A green link "Forgot Password?" is positioned between the fields. Below the fields is a green "Login" button. At the bottom of the form is a green link "Don't have an account? Register here". The screen has a light pink background and a black Android navigation bar at the bottom.

- **Registration**

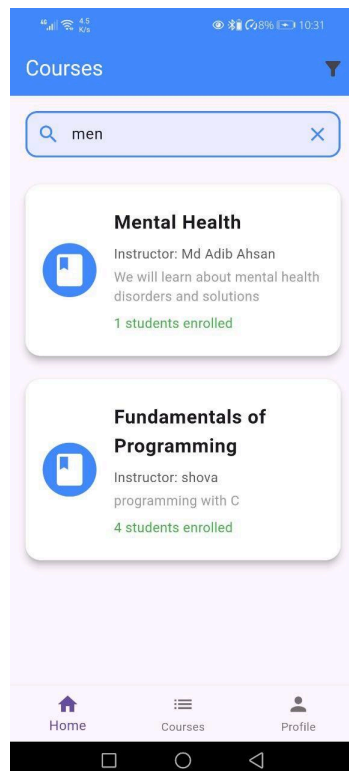


A mobile app screen for student registration. At the top is a blue header bar with a white back arrow and the text "Student Registration". The main content is a light purple rounded rectangle with a shadow. It contains four input fields stacked vertically: "Full Name" with a person icon, "Email" with an envelope icon, "Password" with a lock icon, and "Grade Level" with a graduation cap icon. Below these fields is a blue "Register" button. The screen has a light pink background and a black Android navigation bar at the bottom.

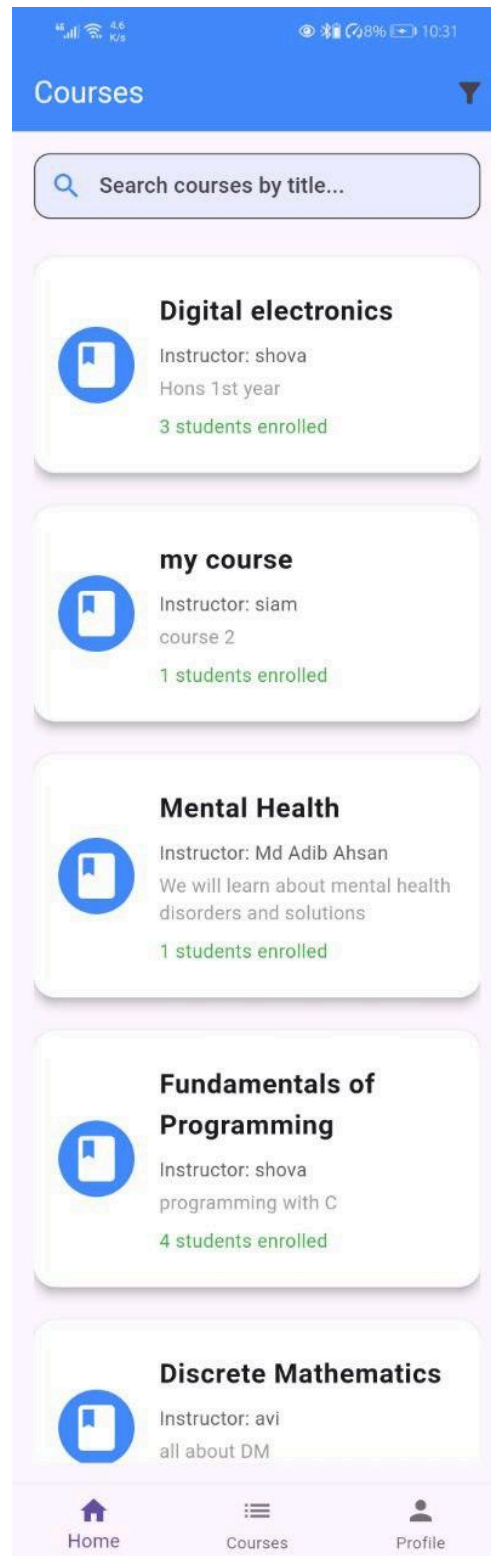
- **Forgot Password**



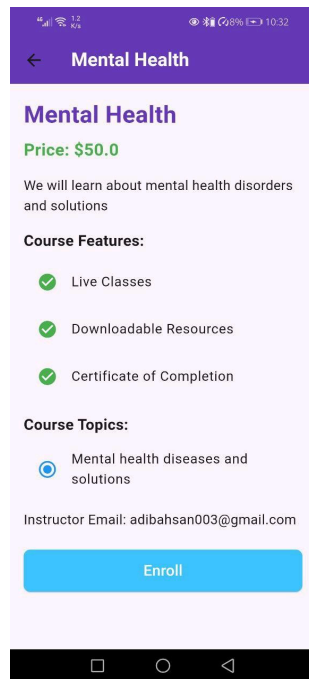
- **Search Course**



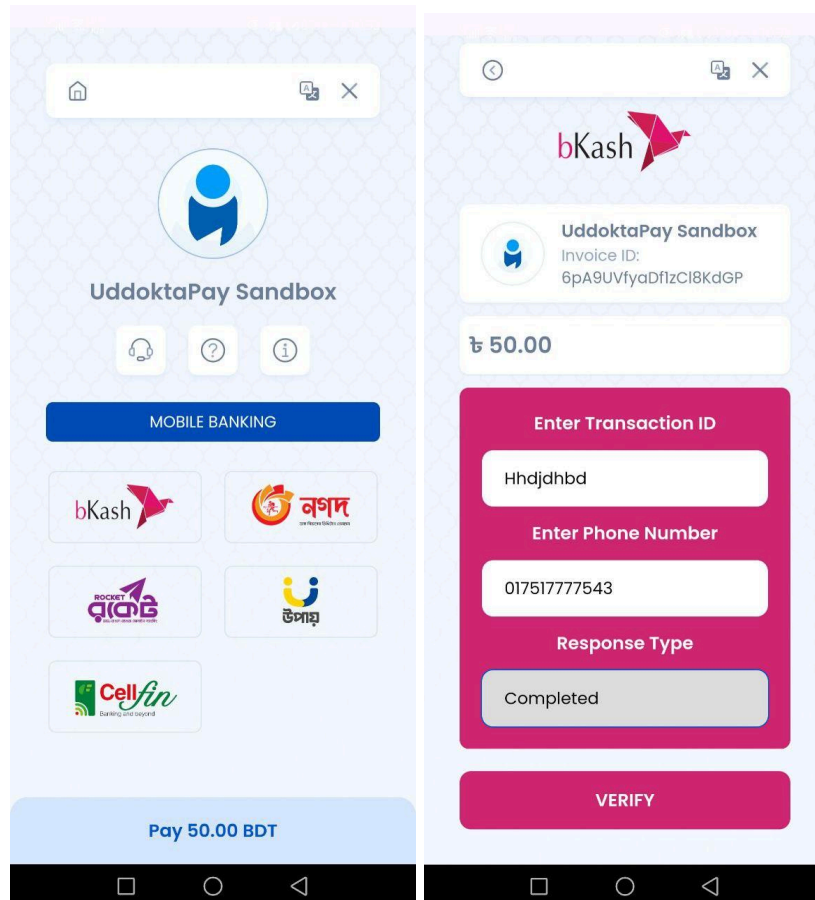
○ All available courses Page



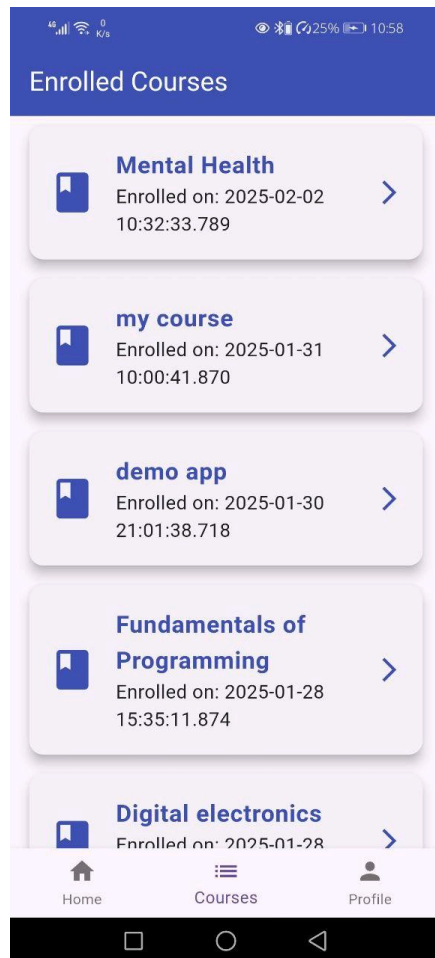
○ **Enroll course**



○ **Payment Gateway**



- **Enrolled Courses**



○ Course

Course Details

Fundamentals of Programming

programming with C

Available Classes

+

Variable

Scheduled At: 9

+

Data type

Scheduled At: 11

Quizzes

?

quizNew

Created At: 2025-02-01 11:56:01.161

?

pointer

Created At: 2025-01-28 15:32:43.241

?

stack

Created At: 2025-01-28 15:33:44.352

?

data type

Created At: 2025-01-28 15:30:42.781

?

introduction

Created At: 2025-01-28 15:26:57.630

Course Documents

IMG_20250128_152740.jpg

Uploaded: 2025-01-28 15:28:01.540579

↓

IMG_20250128_152811.jpg

Uploaded: 2025-01-28 15:29:11.170125

👁

renaming.jpg

Uploaded: 2025-02-01 11:57:04.800100

↓

Leaderboard

Chat

- Available Classes

Available Classes

 **Variable**
Scheduled At: 9

 **Data type**
Scheduled At: 11

- Available Quiz

Quizzes

 **quizNew**
Created At: 2025-02-01
11:56:01.161

 **pointer**
Created At: 2025-01-28
15:32:43.241

 **stack**
Created At: 2025-01-28
15:33:44.352

 **data type**
Created At: 2025-01-28
15:30:42.781

 **introduction**
Created At: 2025-01-28
15:26:57.630

- Available Documents

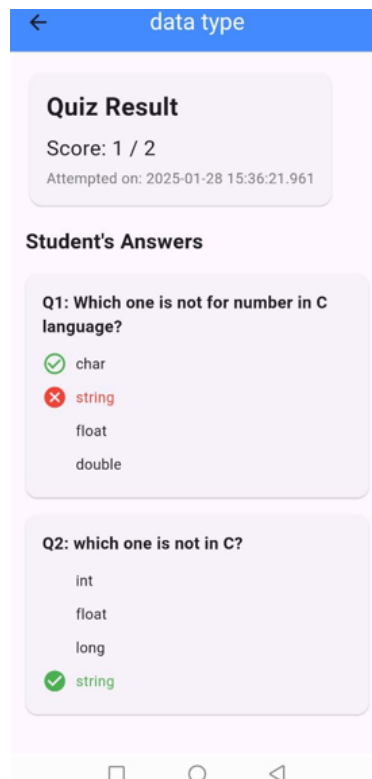
Course Documents

 IMG_20250128_152740.jpg
Uploaded: 2025-01-28
15:28:01.540579 

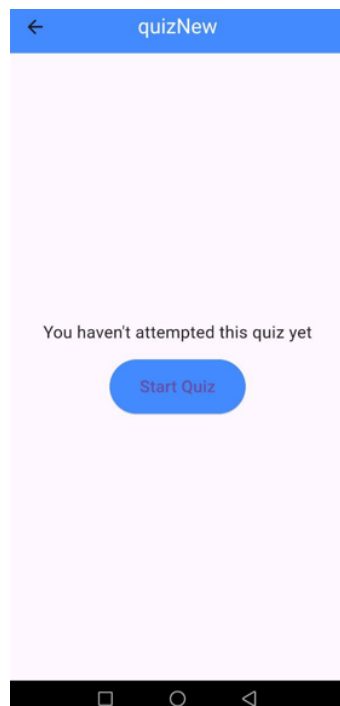
 IMG_20250128_152811.jpg
Uploaded: 2025-01-28
15:29:11.170125 

 renaming.jpg
Uploaded: 2025-02-01
11:57:04.800100 

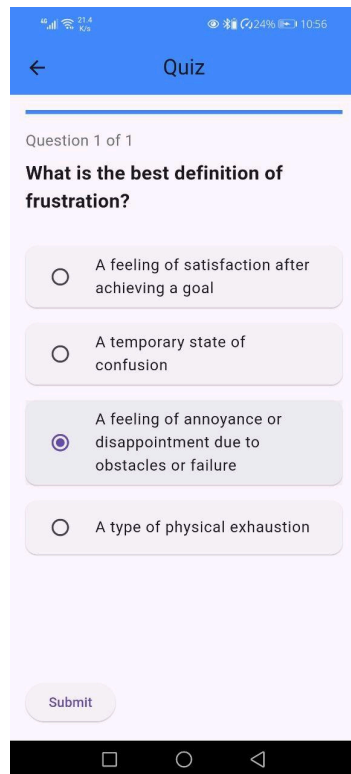
○ **View Attempted Quiz**



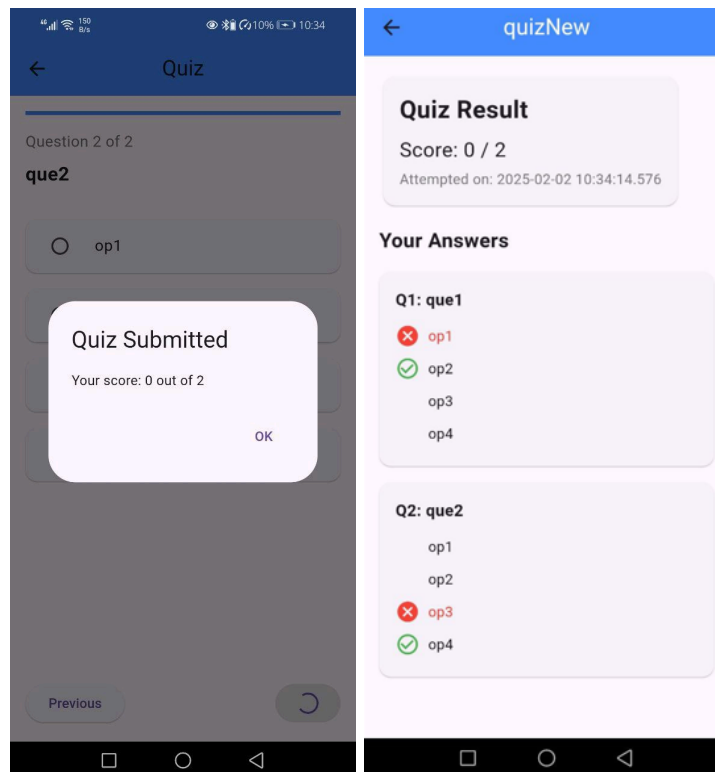
○ **New Quiz**



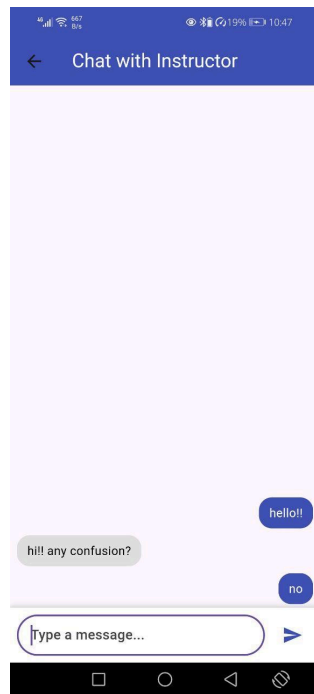
○ **Attempt Quiz**



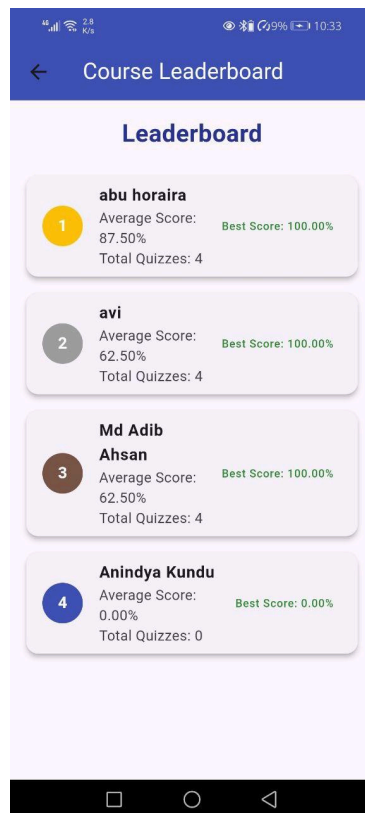
○ **Submit and Result**



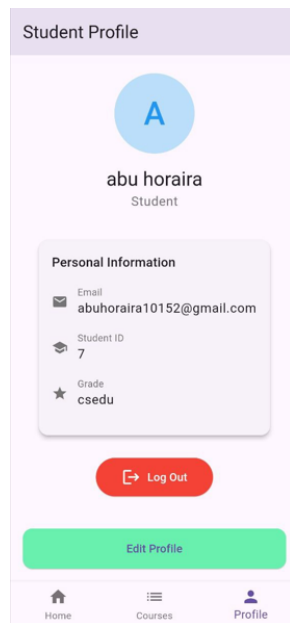
- Chat with Instructor



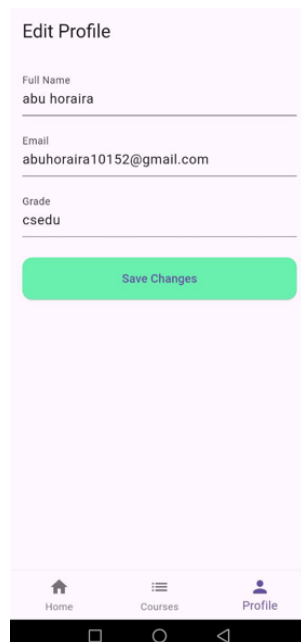
- Leaderboard



- **View Document Profile**

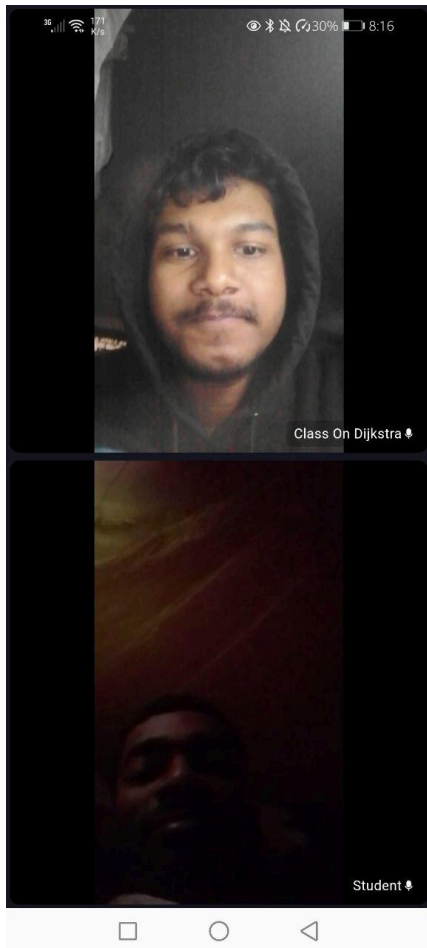


- **Edit Profile**

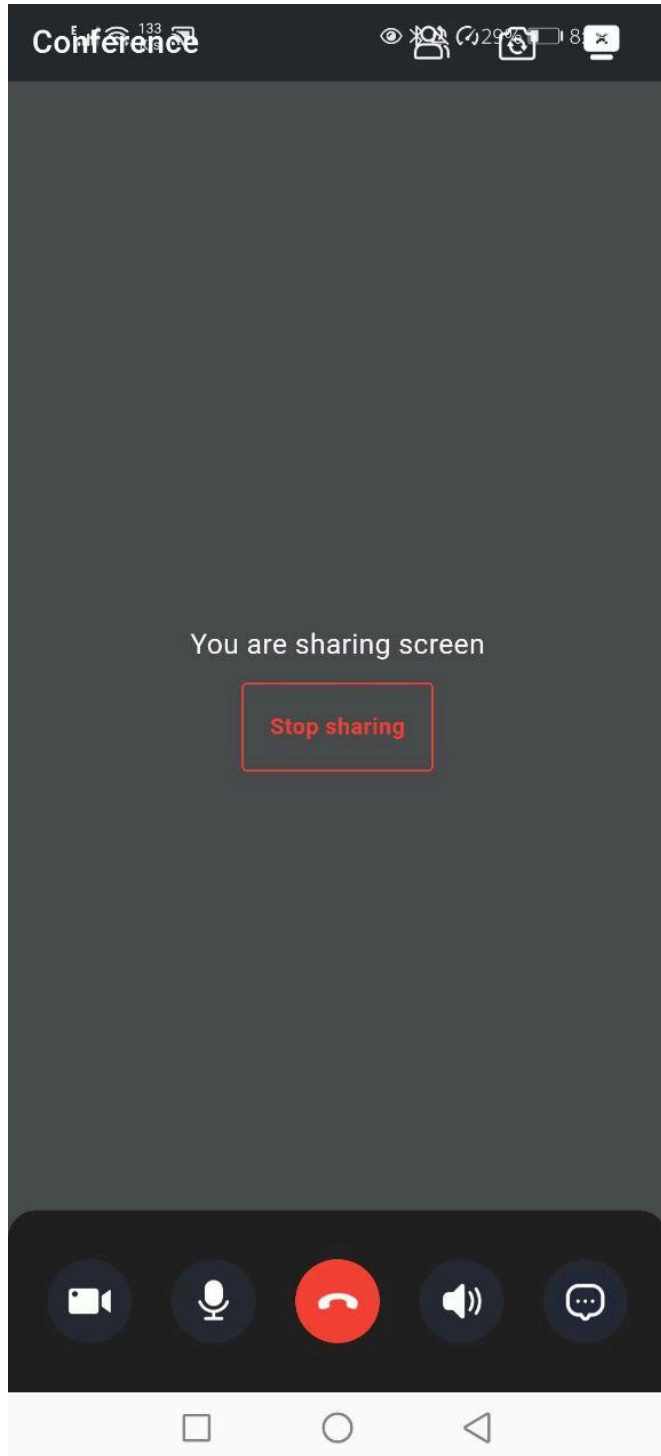


Live Class

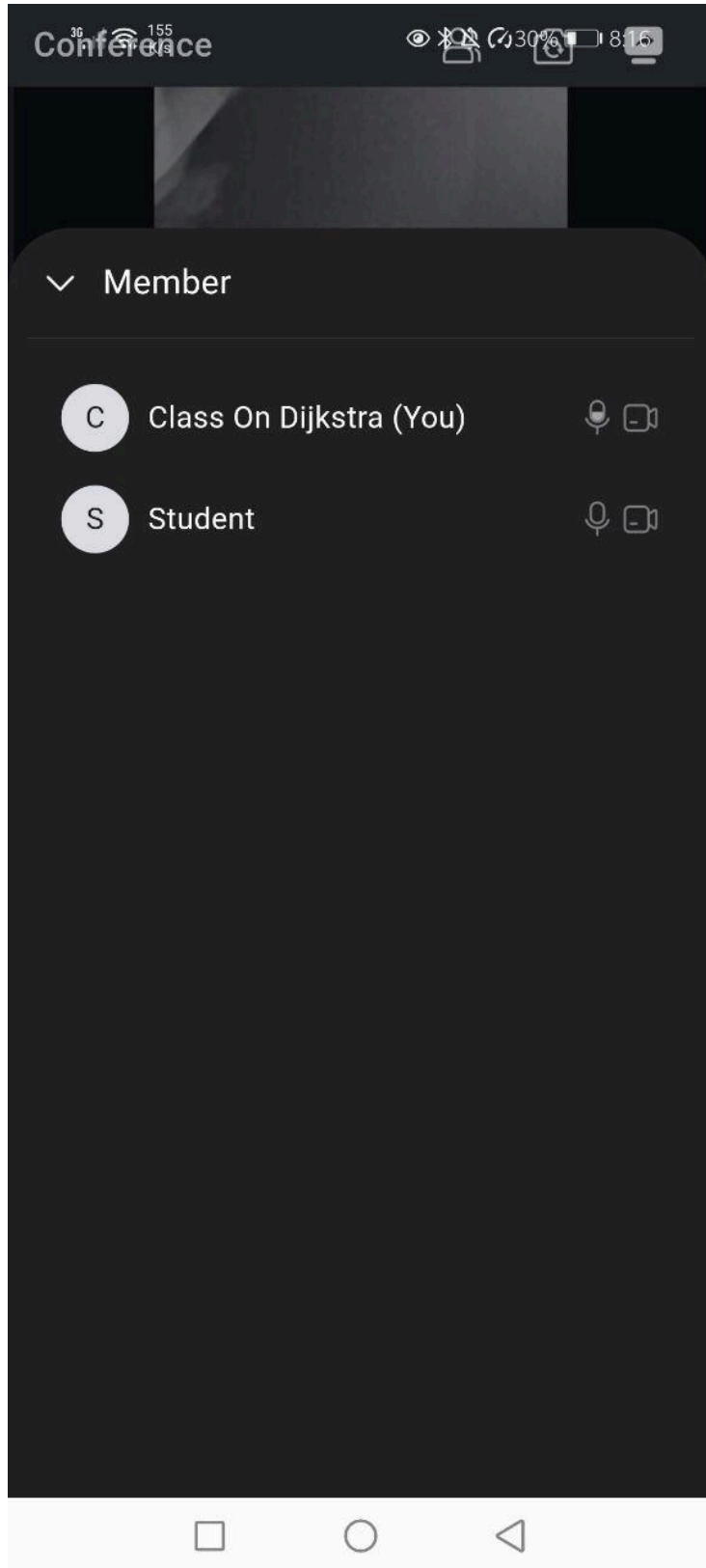
- see all the participants
- supports chatting
- screen sharing
- camera switch
- unmute option
- camera off option



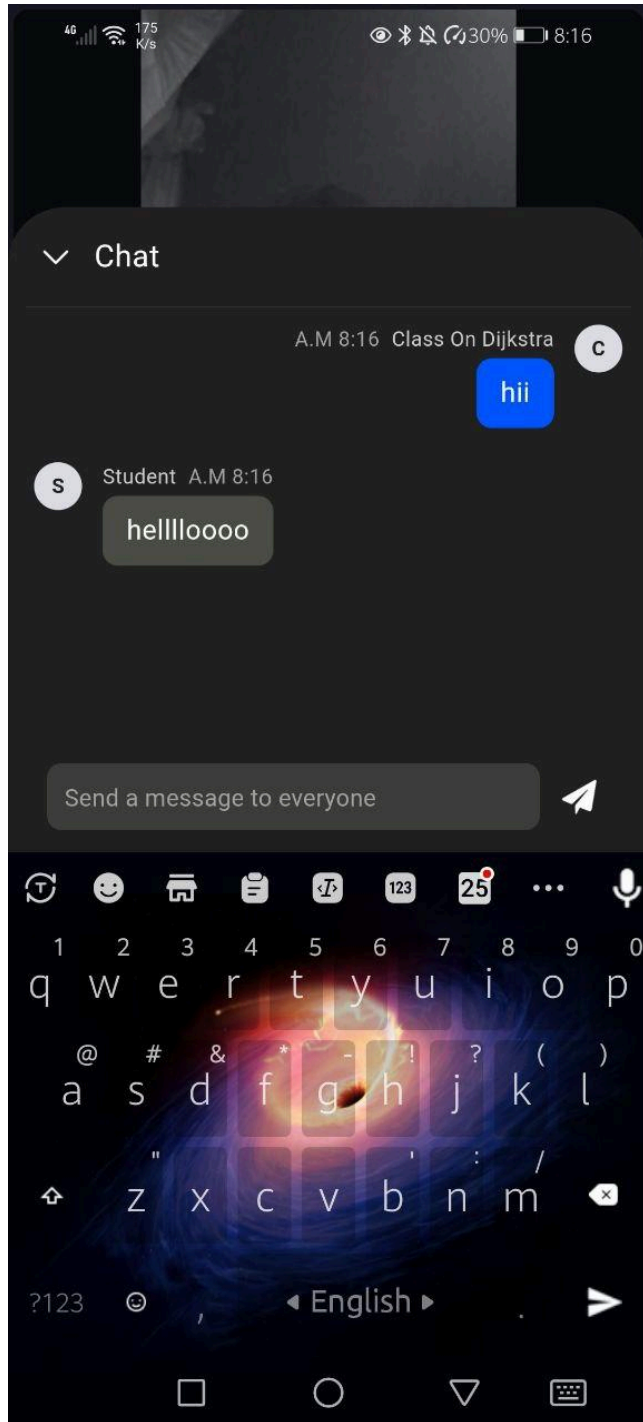
Screen Share



Members:



Chat



2.2 Database Schema

Collections and their attributes:

Users:

1. displayName
2. email
3. expertise / grade
4. role
5. uid (primary key)
6. userId

Courses:

1. courseId (primary key)
2. createdAt
3. description
4. enrolledStudents {}
5. features {}
6. files {}
7. price
8. title
9. topics {}

Enrollment:

1. courseId
2. studentId
3. title
4. enrolledAt
5. quizzes{}
Composite Primary Key [courseId + studentId]

Classes:

1. classDescription
2. classTime
3. classTitle (Primary Key)
4. conferenceId
5. courseID
6. createdAt
7. enrolledStudent{}
8. instructorId
9. maxStudents

Quizzes:

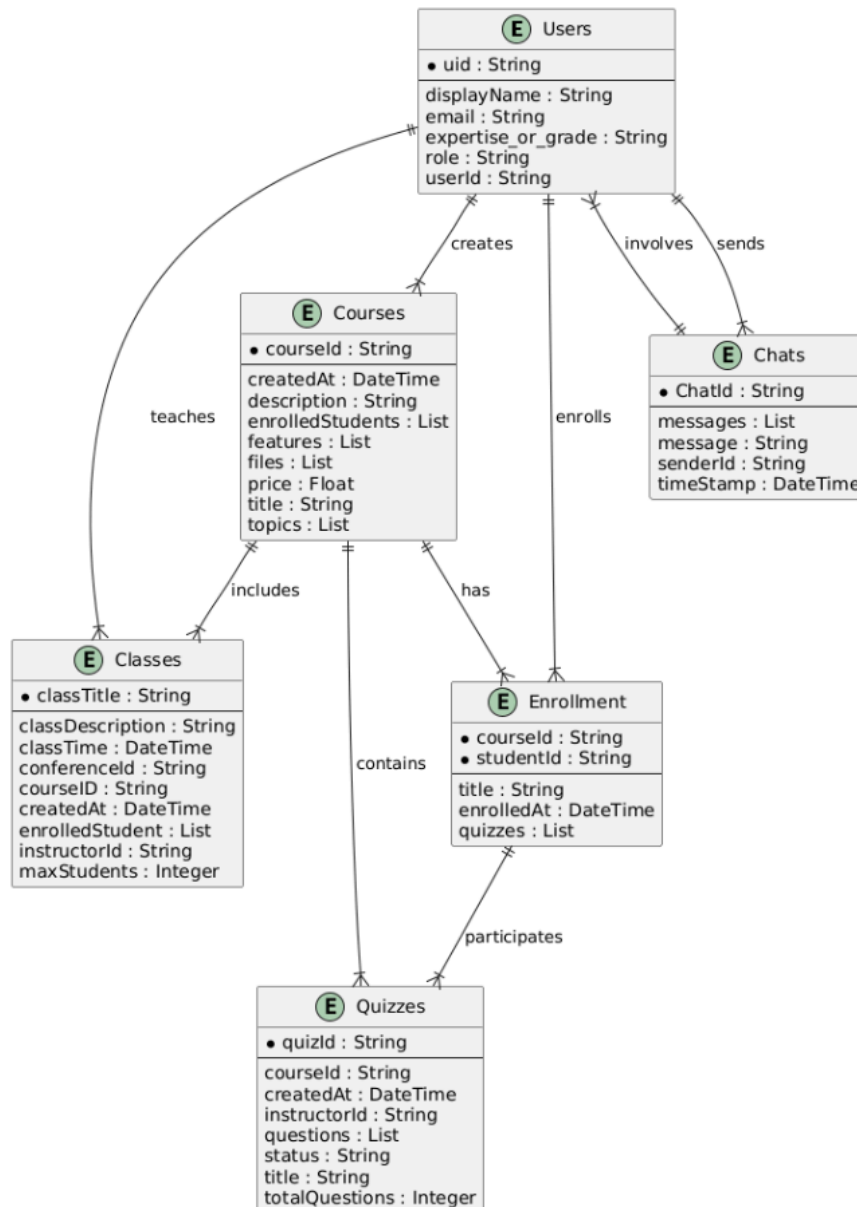
1. courseId
2. createdAt
3. instructorId
4. questions{}
5. quizId (Primary Key)

6. status
7. title
8. totalQuestions

Chats:

1. ChatId(Primary Key)
 - a. messages
 - i. message
 - ii. senderId
 - iii. timeStamp

DIAGRAM



2.3 Code Repository Link

[View gitHub Repository](#)

2.4 Video Demo Link

[Drive Link For Video](#)
[Youtube Link](#)

3. Conclusion

3.1 Challenges and Solutions

Challenge 1: Frequent updates

Issue:

Our project used Flutter for mobile development. Libraries are updating in every short period

Impact:

In a Flutter project, dependencies (packages and libraries) frequently update, which can result in Compatibility issues between dependencies. Increases development time and effort to resolve compatibility issues.

Solution:

flutter pub upgrade

Challenge 2: Go Router Using Issue

Issue:

When we used go router there were some issue for saving user details once logged in

Impact:

Automatic login(after login for the first time) was not happening because of ambiguity.

Solution We Attempted:

Switched the go router statements to Navigator.push() and Navigator.pop()

Challenge 3: Firebase Rules Upgrade

Issue:

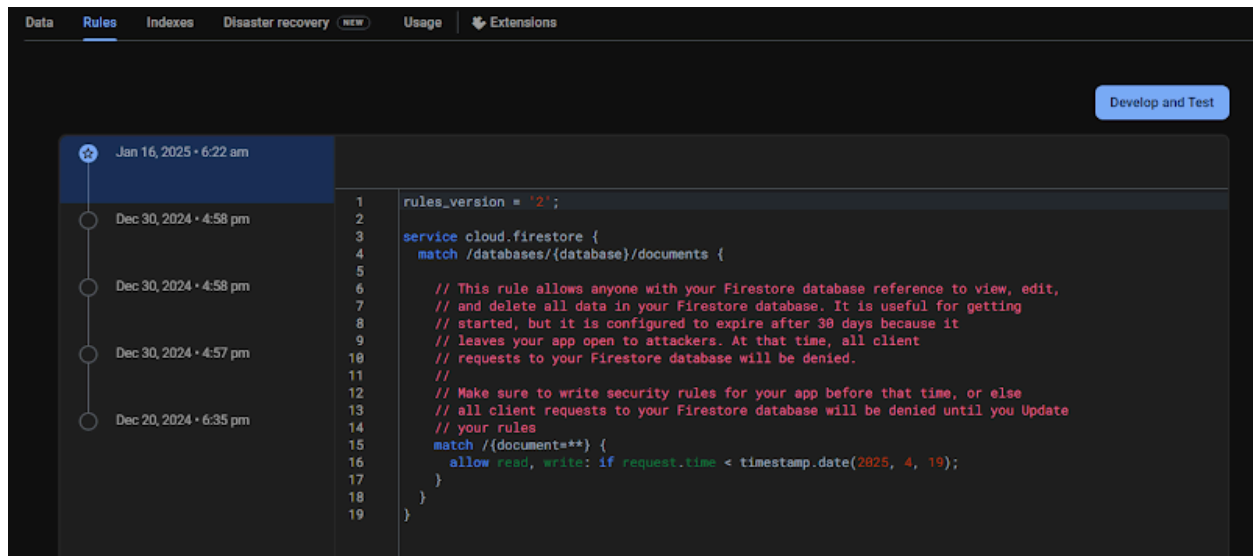
“Firestore rules” is a restriction for who, when, and how to update the firebase database and storage. By default it is one month from the creation of the database and for the users

Impact:

We were unable to insert data.

Solution:

Update the Rules



Challenge 4: Firebase Storage using

Issue:

Unable to upload photos, pdfs, videos.

Impact:

Before June 2024 it was easy to upload documents on firebase storage(firebase). Now it needs a credit or debit card for verification

Solution:

Adding a debit card will solve this problem

Challenge 5: Build Gradle Issues with

`zego_uikit_prebuilt_video_conference`

Issue: When integrating the `zego_uikit_prebuilt_video_conference` package into our Flutter project, we encountered errors during the build process. The build failed due to incompatible dependencies and incorrect configurations in the `build.gradle` files.

Specifically, there were version conflicts related to the ZegoExpressEngine SDK, causing errors like `Could not find method implementation() for arguments` and `API level mismatch`.

Impact:

The project failed to build or run on Android devices.

The error prevented us from utilizing video conference features in the app.

Solution:

We updated the `gradle-wrapper.properties` file to use the latest compatible version of Gradle and the Android Gradle plugin. This resolved issues related to outdated build tools.

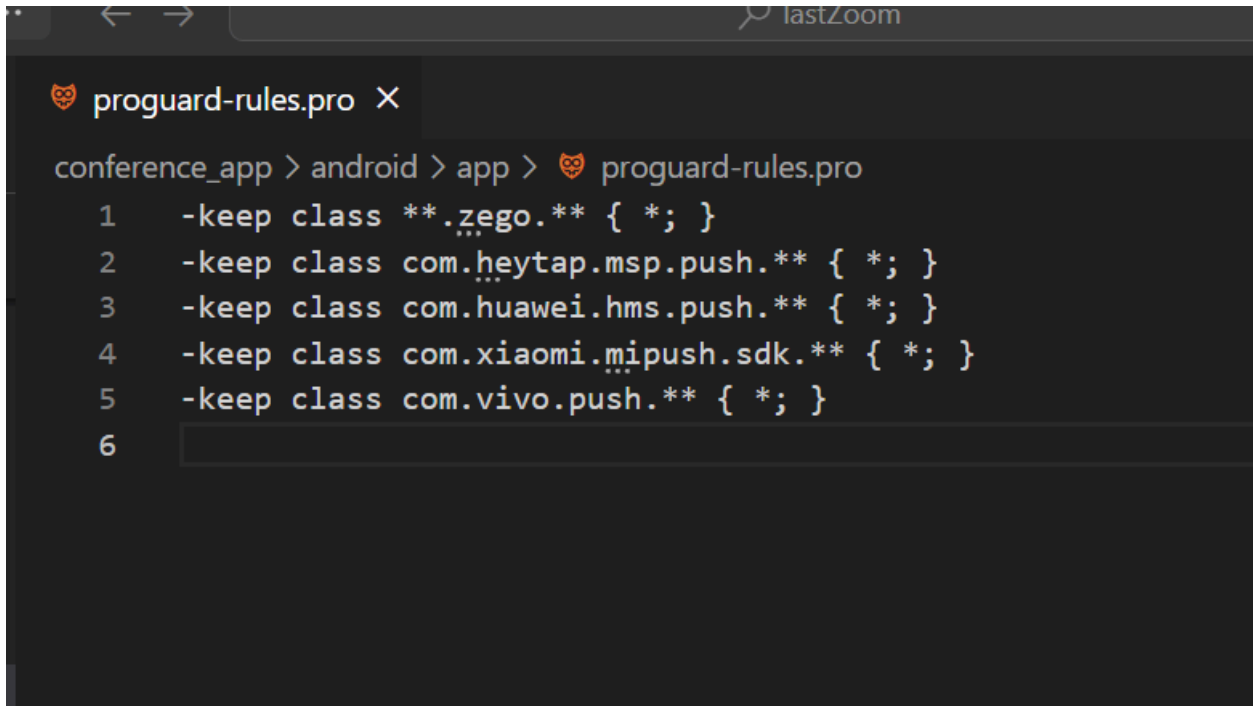
```
Welcome  gradle-wrapper.properties X
conference_app > android > gradle > wrapper > gradle-wrapper.properties
1  distributionBase=GRADLE_USER_HOME
2  distributionPath=wrapper/dists
3  distributionUrl=https\://services.gradle.org/distributions/gradle-8.3-bin.zip
4  networkTimeout=10000
5  validateDistributionUrl=true
6  zipStoreBase=GRADLE_USER_HOME
7  zipStorePath=wrapper/dists
8
```

We checked and ensured that the target SDK and compile SDK versions in android/app/build.gradle were set to compatible levels for the package. This involved aligning the minSdkVersion and targetSdkVersion to match the requirements of the package.

```
Welcome  gradle-wrapper.properties  build.gradle X
conference_app > android > app > build.gradle > android
11  android {
12      //
13      //
14      //
15      //
16      //
17      //
18      //
19      //
20      //
21      kotlinOptions {
22          jvmTarget = JavaVersion.VERSION_1_8
23      }
24      //
25      defaultConfig {
26          //
27          applicationId = "com.example.edtech"
28          //
29          minSdk = 23
30          targetSdk = flutter.targetSdkVersion
31          versionCode = flutter.versionCode
32          versionName = flutter.versionName
33      }
34      //
35      buildTypes {
36          release {
37              signingConfig = signingConfigs.debug
38              minifyEnabled false
39              shrinkResources false
40              proguardFiles getDefaultProguardFile('proguard-android.txt'), 'proguard-rules.pro'
41          }
42      }
43  }
44
```

We have made a proguard-rules.pro file under android/app and added specific ProGuard

rules to keep the necessary classes and methods intact for ZegoUIKit. This was done by adding the following rules to the `proguard-rules.pro` file:



```
conference_app > android > app > proguard-rules.pro
1  -keep class **.zego.** { *; }
2  -keep class com.heytao.msp.push.** { *; }
3  -keep class com.huawei.hms.push.** { *; }
4  -keep class com.xiaomi.mipush.sdk.** { *; }
5  -keep class com.vivo.push.** { *; }
6
```

Challenge 6: Issue with zego_uikit_prebuilt_video_conference Integration

Issue: While integrating the `zego_uikit_prebuilt_video_conference: ^2.9.8` package for video calling functionality, we encountered an issue where the user could not join the conference room due to missing or incorrect user data.

Impact:

Users could not join the video conference call due to missing user data.

App experienced crashes or failed to load the video call page because `null` values were passed to the `ZegoUIKitPrebuiltCall` widget.

Solution:

Ensured that Firebase authentication data (`userID`, `userName`) was fetched asynchronously before initiating the video call.

We refactored the code to use `async/await` properly, ensuring user data was fetched from Firebase before initializing the video call.

Challenge 7: Url Launcher problem

Issue: The `launchUrl` or `canLaunchUrl` method always returns false or the URL doesn't open.

Impact: gmail app is not opening

Solution:

Give permission: `<uses-permission android:name="android.permission.INTERNET" />`

Challenge 8: User permissions

Issue: App need some permission in the running devices

Impact: App is not running or can't access required files

Solution:

```
<uses-permission android:name="android.permission.QUERY_ALL_PACKAGES" />
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
<uses-permission android:name="android.permission.ACCESS_WIFI_STATE" />
<uses-permission android:name="android.permission.RECORD_AUDIO" />
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
<uses-permission android:name="android.permission.CAMERA" />
<uses-permission android:name="android.permission.BLUETOOTH" />
<uses-permission android:name="android.permission.MODIFY_AUDIO_SETTINGS" />
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
<uses-permission android:name="android.permission.READ_PHONE_STATE" />
<uses-permission android:name="android.permission.WAKE_LOCK" />
```

```
instructor\profile.dart  AndroidManifest.xml ×
1  <manifest xmlns:android="http://schemas.android.com/apk/res/android">
2    <uses-permission android:name="android.permission.QUERY_ALL_PACKAGES" />
3    <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
4    <uses-permission android:name="android.permission.ACCESS_WIFI_STATE" />
5    <uses-permission android:name="android.permission.RECORD_AUDIO" />
6    <uses-permission android:name="android.permission.INTERNET" />
7    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
8    <uses-permission android:name="android.permission.CAMERA" />
9    <uses-permission android:name="android.permission.BLUETOOTH" />
10   <uses-permission android:name="android.permission.MODIFY_AUDIO_SETTINGS" />
11   <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
12   <uses-permission android:name="android.permission.READ_PHONE_STATE" />
13   <uses-permission android:name="android.permission.WAKE_LOCK" />
14   <application
```

Challenge 9: MinSdk Update

Issue: sdk failure

Impact: some programmes are not running for old versions. Like Sdk version 21

Solution: update minSDK

```
defaultConfig {
    // TODO: Specify your own unique Application ID
    (https://developer.android.com/studio/build/application-id.html).
    applicationId = "com.example.edtech"
    // You can update the following values to match your application needs.
    // For more information, see: https://flutter.dev/to/review-gradle-config.
    minSdk = 23
    targetSdk = flutter.targetSdkVersion
    versionCode = flutter.versionCode
    versionName = flutter.versionName
}

defaultConfig {
    // TODO: Specify your own unique Application ID (https://developer.android.com/studio/build/application-id.html).
    applicationId = "com.example.edtech"
    // You can update the following values to match your application needs.
    // For more information, see: https://flutter.dev/to/review-gradle-config.
    minSdk = 23
    targetSdk = flutter.targetSdkVersion
    versionCode = flutter.versionCode
    versionName = flutter.versionName
}
```

Challenge 10: Assets file

Issue: file not found

Impact: Pictures or images not showing

Solution: ensure images folder is in pubspec.yaml

```
assets:  
  - assets/  
  
flutter:  
  
  # The following line ensures that the Material Icons font is  
  # included with your application, so that you can use the icons in  
  # the material Icons class.  
  uses-material-design: true  
  assets:  
    - assets/
```

Challenge 11: Document Upload Failed

Issue: Document is not uploading

Impact: Document feature is not working properly.

Solution: Using free databases. We used Firestore which is paid (1 month free limit)

Challenge 12: Live class

Issue: live class will not work after 1 month free access

Impact: Live class failure

Solution: Updating live class userid to use free.

3.2 Lessons Learned

- **Understanding User Needs:** We realized the importance of designing an intuitive UI/UX for better user engagement.
- **Using GitHub:** Learning github uses was one of the major lessons of this project. We used github a lot for developing the project. At the beginning some of our commits were done by one account and the issue was fixed later.
- **Using open source:** Solution of many problems was done by open sources like [StackOverFlow](#). Also some help was taken from chatgpt and AI for more efficient workflow.
- **Target Fixing/ Deadline:** Fixing date based targets is important for industrial jobs. We targeted many deadlines for some specific features.
- **Development Approach and Bug solving:** App Testing helped us refine the app. This bug solving mentality is a key lesson of this project.
- **Efficient Database Design:** Firestore's NoSQL structure required careful planning to ensure optimized queries and real-time updates.
- **Security Best Practices:** We learned to implement Firebase Authentication securely.
- **Real-Time Functionality:** Implementing live classes and chat reinforced our understanding Firestore, and real-time synchronization.
- **Work Management:** We divided our job particularly for fast and efficient outcome

3.3 Future Plan

The app is currently in its basic version having only the bare minimum features. We'd like to improve the user interface to make it appealing to the users. We'd also like to incorporate useful features like:

- **Revamping the User Interface (UI/UX):** Improving the design to make it more visually appealing and user-friendly.
- **Enhancing Live Class Experience:** Adding screen-sharing, whiteboard features, and session recordings.
- **Gamification Features:** Introducing badges, rewards, and additional leaderboard functionalities to boost engagement.
- **AI-Powered Recommendations:** Implementing AI to suggest courses based on user preferences and performance.
- **Offline Mode:** Enabling users to download course materials and quizzes for offline access.
- **Improved Chat System:** Adding group chat, voice messages, and multimedia sharing for better communication.
- **Multi-Language Support:** Expanding language options to cater to a wider audience.